DS & AI Lab Project (BSDA4001)

TerraWatch: Intelligent Satellite Image Change Detection and Analysis

MILESTONE 1: PROBLEM DEFINITION & LITERATURE REVIEW

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BS in Data Science and Applications

GROUP 7

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1. Introduction and Problem Definition

Satellite image comparison reveals building and road changes. Masks localise differences, while captioning and interactive systems describe them [4-6]. However, descriptions can misstate objects, quantities or locations; lighting, viewpoint and alignment differences complicate interpretation.

Problem statement. How can two images of the same place be converted into a localised, plain-language description of visible urban change, while identifying claims that the available visual evidence does not support?

TerraWatch retains its focus on detection, grounded descriptions and reliability. A caption and checker may agree with the same incorrect mask. We will test whether inspecting both original images, followed by an accept/reject/defer decision, improves factual reliability without suppressing useful information.

2. Scope, Stakeholders and Practical Relevance

In scope: roughly co-registered, high-resolution RGB image pairs; labelled building/road change; masks, descriptions and traceable evidence; nuisance tests; and a local demo. Appearance/removal claims require time-specific evidence. Groundwork requires observable, appropriately labelled examples.

Out of scope: deciding legality, ownership, intent or cause; unverified damage/impact estimates; nationwide or real-time operation; exact geographic measurements without suitable metadata; and validated long-term trajectories from only two observations. Lower-resolution multispectral monitoring is a separate extension.

Stakeholder	Need and proposed value
Municipal building/estate officers; primary	Review visible construction or encroachment candidates using a mask, concise description and uncertainty flag; determine legality through external records and field checks
Urban planners; secondary	Compare visible built-up changes across image pairs
Remote-sensing researchers and non-specialist users; secondary	Inspect reproducible outputs and understand what the images do and do not establish

The intended value is better review support. Reduced officer workload is a potential benefit, not an established outcome; demonstrating it would require a separate user study.

3. Objectives

O1: train and evaluate a credible change-detection baseline. **O2:** assess sensitivity to nuisance variation and domain shift. **O3:** test whether image-backed claim checking improves factual correctness while retaining informative content. **O4:** deliver a reproducible local pipeline with measured runtime and memory. Section 9 defines how these objectives will be assessed.

4. Literature Review: Study-Comparison Matrix

The matrix separates mechanisms from evaluation. A paper's use of conventional caption scores does not establish an absence of grounding or verification.

Study	Objective	Datasets / setting	Experimental design	Main finding	Limitation relevant to TerraWatch
FC-Siam [1]; Reality Check [3]	Establish effective detection baselines	Urban change benchmarks	Early fusion/Siamese networks; controlled U-Net comparisons	Simple CNN baselines remain important	Detection scores alone do not validate a description
ChangeFormer [2]	Model multiscale change context	LEVIR-CD and DSIFN-CD	Siamese hierarchical transformer; detector comparisons	Strong published detection baseline	Label/domain dependence; no standalone claim verifier
Chg2Cap [7]; RSCaMa [8]	Describe image-pair changes	LEVIR-CC	Attentive captioning; spatial/temporal state-space features	Specialist change representations support captioning	Caption similarity does not fully establish factual correctness
Change-Agent [4]	Interactive comprehensive change analysis	LEVIR-MCI	Joint mask/caption perception, feature interaction and LLM tool use; ablations	Integrates pixel, text and object-statistic outputs	Shared perception errors can propagate; evaluate checking under faulty evidence
ChangeChat [5]	Interactive change questions	LEVIR-derived instructions (87k)	Rule/GPT-assisted instruction tuning; caption, count and localisation tasks	Supports task-specific interactive analysis	Generated supervision and shared imagery require provenance and error audits
CCExpert [6]	Improve MLLM change understanding	LEVIR-CC; CC-Foundation	Difference-aware integration, multiscale features and staged training	Explicit difference features improve reported caption metrics	Grounding is already prior art; gains do not establish calibration under domain shift
KCFI [9]; TEOChat [10]	Detection-guided captioning; temporal EO dialogue	LEVIR-CC; multi-task EO instruction data	Pixel-task-constrained features; temporal instruction tuning	Visual guidance and temporal multi-task modeling are established	Broad capabilities do not substitute for a controlled claim-checking experiment
Ricci et al. [11]	Zero-shot descriptions and fact matching	SECOND; LEVIR-CC	Pretrained LLM/VLM approaches; FMScore	Fact-oriented evaluation already exists in this task	Reference-fact matching and inference-time visual verification answer different questions
Spot-the-shift [12]	Evaluate grounded difference descriptions	Street-view structural-change pairs	Human-verified change units, spatial annotations and no-change evaluation	Atomic factual/coverage evaluation is established	Different domain; its principles inform, rather than establish novelty for, our protocol
TerraScope [26]	Grounded EO reasoning	EO reasoning; building change estimation	Visual reasoning coupled with segmentation; task comparisons	Pixel grounding is an established reasoning approach	Grounded reasoning alone is not novel; test the additional claim-decision policy

Published reference points. Change-Agent reports 86.43 mIoU, 65.95 BLEU-4 and 140.29 CIDEr-D for its MCI model on LEVIR-MCI [4, Table II]. Within that table, Chg2Cap reaches 64.39 BLEU-4 and 136.61 CIDEr-D. These are literature results, not TerraWatch measurements or targets. Cross-paper scores are not treated as directly comparable without matching splits and evaluation scripts.

Current practice. Image differencing offers a transparent diagnostic but confounds appearance differences with change. Learned detectors improve spatial prediction; captioners and agents add semantic interpretation. The remaining question is how reliably an output can be checked when its own evidence source is wrong. Generic models' capability claims are not evidence of satellite-domain accuracy.

5. Research Gap and Proposed Contribution

5.1 What prior work already addresses

Integration and tool use are established in Change-Agent; interactive counting/localisation in ChangeChat; difference-aware visual conditioning in CCExpert and KCFI; and factual evaluation in FMScore and grounded change-unit evaluation [4-6,9,11,12]. Modularity, mask conditioning and human evaluation are therefore design choices or established practices, not independent novelty claims.

5.2 Specific mechanism to investigate

We propose **disagreement-triggered visual reinspection**. Each description is decomposed into claims linked to candidate regions. Consistency, visibility and uncertainty checks identify conflicts or insufficient evidence. Triggered claims are queried on corresponding crops from both original images, with full-image context retained; a validation-designed audit of untriggered claims measures missed errors. A separate decision layer uses these observations, quality flags and mask evidence to accept, reject or defer each claim. Deferred claims are qualified or withheld, with the reason preserved.

A binary mask cannot establish object identity or addition/removal direction. Region counts are not building counts. The decision layer therefore cannot approve a claim solely because the caption matches the mask. Its thresholds are selected on validation data, and its visual observations remain fallible predictions.

5.3 Mechanism-level comparison

System	How visual evidence informs output	Specific distinction examined here
Change-Agent [4]	Joint detection/caption features and agent tools	Hold perception fixed and test acceptance of claims when that perception is faulty
ChangeChat [5]	Instruction-tuned paired-image analysis	Compare free-form answers with explicit claim decisions and deferral
CCExpert [6]	Difference-aware features integrated before language decoding	Compare generation-time grounding with additional targeted reinspection after generation
TerraWatch; proposed	Claim-to-region links, disagreement triggers, original-image reinspection and bounded decisions	Test whether reinspection adds information beyond mask consistency at useful coverage

This comparison does not assert that prior systems lack all checking. The experimental distinction is the explicitly isolated reinspection-and-decision policy and its response to perception errors.

5.4 Contribution and supporting evidence

Primary hypothesis: image-backed checking yields better human-judged factual precision and reference-fact coverage than mask-only checking on identical candidate descriptions, particularly for naturally incorrect masks. Targeted crops offer evidence unavailable to a mask-only checker; constrained decisions allow uncertainty to be represented rather than converted into confident prose. Prior work supports these ingredients, but their benefit in this proposed configuration must be demonstrated.

The intended contribution is a reliability mechanism evaluated through a detector-error-aware protocol. It is currently a proposed adaptation and empirical investigation, not an established first-of-kind architecture. A meaningful advance requires improvement over mask-only checking and a simple image-backed judge, while controlling retained content, compute and domain shift. If these comparisons show no gain, the contribution will be reported as an empirical analysis of failure modes.

6. Proposed Approach and Technical Grounding

The existing detector-to-description pipeline is retained. Four learned roles are considered; preprocessing, schema validation and deterministic checks remain explicit software components.

Image pair → quality/scale checks → change detector → evidence-linked caption → visual reinspection → accept/reject/defer → report.

Role	Candidate and training strategy	Output
Change perception	Compact early-fusion/Siamese U-Net trained as a baseline; ChangeFormer as a stronger comparison [1-3]	Change probabilities, class masks where supervised, candidate regions
Change description	Specialist baseline such as Chg2Cap/RSCaMa; Qwen3-VL as a candidate for multi-image prompting or adapter tuning [7,8,20]	Short caption and structured claim spans
Visual evidence inspection	Frozen VLM initially; inspect matched T1/T2 crops and context	Object, action, location observations and visibility/ambiguity flags
Decision policy	Rules baseline; Clef/Clef-flash or another bounded classifier; Jev as an optional comparator [21-23]	Claim acceptance, contradiction or deferral and logged decision scores

Input adaptation. Inspect dimensions, channels, metadata, blur, saturation and alignment. Apply model-required normalisation, paired tiling and coordinate-preserving resizing. Use declared ground-sampling distance when available; do not infer physical resolution from image dimensions. Radiometric adjustment is optional and compared against unchanged inputs because histogram matching can remove genuine change. Retain raw images and transformation metadata; reject incompatible modalities rather than attempting universal automatic correction.

Learning plan. Train a compact detector and compare BCE-plus-Dice with a focal-loss alternative on audited class distributions. Use synchronised spatial augmentations and controlled one-date appearance perturbations. Fine-tune only selected caption/visual adapters if a frozen baseline proves inadequate and compute permits. The visual inspector may initially reuse frozen VLM weights; a different backbone is an ablation, not an assumption of independent errors. A decision model is retained only if it improves held-out decisions over rules and a general VLM judge.

Decision-model rationale. Clef is an open decision-model family; current Ollama listings include multimodal variants [21,22]. Jev's October release introduces experimental image support [23]. These make bounded claim decisions feasible candidates, but structured responses and confidence scores do not prove factual correctness. Recent Jev evaluation identifies wording/order sensitivity [24]; we will test calibration and decision stability on this task. No remote-sensing reliability gain is assumed from unrelated vendor benchmarks.

Reporting safeguards. Preserve original/revised captions, evidence links and deferral reasons. Exact counts require validated instance evidence; geographic area requires suitable spatial metadata. A crop's upper-right is image-relative, not geographic northeast unless orientation is known. Predicted evidence and human reference annotations remain separate.

7. Data Availability, Alignment and Robustness Strategy

LEVIR-MCI is the preferred primary dataset because it extends LEVIR-CC with aligned change masks; the shared family has 10,077 image pairs and five captions per pair [4,13]. Dataset access, label dictionaries and asset-specific usage terms will be audited before training. Source links have been identified; complete payload/rights validation is not claimed.

The broader plan considers a portfolio of nine releases. It is not nine independent populations and will not be concatenated indiscriminately.

Data group	Candidate releases and supervision	Use / boundary
Aligned detection and text	LEVIR-MCI, LEVIR-CC [4,13]	Core training/evaluation; same underlying pairs counted once
Building change	LEVIR-CD, LEVIR-CD+, S2Looking [14,15]	Compatible detection supervision after overlap audit; hold out a domain for transfer
Broader visual change	SYSU-CD [16]	Separate transfer evaluation; reconcile broader label coverage before joint training
Semantic transitions	SECOND [17]	Optional task-specific head; not interchangeable with binary building labels
Change instructions	ChangeChat-105k [18]	Optional adapter data; trace LEVIR-derived images and instruction provenance
Lower-resolution urban change	OSCD [19]	Separate multispectral extension; not pooled with high-resolution RGB as equivalent building evidence

Combination strategy. Harmonise image ordering, modality, units, label ontology and parent-scene IDs first. A shared backbone may use task/domain-specific adapters or heads. Apply losses only where the corresponding annotation exists; an absent caption or semantic label is not a negative label. Sample domains explicitly to prevent large datasets dominating. Compare single-domain training, compatible mixed training and mixed training plus domain-aware adaptation. More data does not guarantee robustness to unseen imagery.

Leakage control. Preserve official train/validation/test membership and group related crops by parent scene where possible. Check hashes, near duplicates and shared provenance across releases; keep all captions/instructions of a pair together. Reserve an entire compatible dataset/domain for transfer testing rather than training on every source. Track pretrained checkpoint exposure to LEVIR-derived images.

POC limit. Use 20% of unique pairs by default and no more than 30% of any selected dataset, cumulatively across development, training, annotation and evaluation. Shared-image releases do not multiply independent samples. Prefer pretrained inference and short learning pilots; full training belongs to the subsequent project phase.

8. Feasibility and Key Risks

Start with one dataset, compact detection and one description baseline; demonstrate 5-10 pairs end to end before expanding. Select image size, batch size, quantisation and adapters after measuring device memory and runtime. Large multimodal models are optional, not prerequisites. Open/local baselines keep the core reproducible without a paid service.

Main risks are shared detector/verifier errors, ambiguous/touching buildings, incompatible labels, duplicate scenes, destructive normalisation and excessive model cost. Mitigations are original-image inspection, uncertainty/deferral, label-aware heads, provenance audits, preprocessing ablations and staged model selection. Real seasonal/viewpoint robustness requires suitable held-out examples; synthetic corruptions alone cannot establish it.

9. Evaluation and Definition of Success

Use official splits, validation-only tuning, frozen test protocols and identical input pairs across comparisons. Report uncertainty by paired resampling at scene level where possible, otherwise at pair level with the dependence limitation stated. Published scores provide context, not fair full-data targets for a small POC.

Objective	Measures	Evidence of success
O1: detection	Change-class precision/recall, F1 and IoU; class mIoU where applicable	Reproducible learned baseline and fair comparison on the same split; inspect small-object and no-change failures
O2: robustness	Clean-to-corrupted F1/claim-quality drop; separate held-out-domain scores	Smaller degradation than the matched baseline without material clean-quality loss; margins fixed from validation before test
O3: reliable descriptions; primary	Human supported-claim precision, reference-fact recall and their harmonic mean; precision at matched report coverage	Improvement over mask-only checking with retained information controlled; paired confidence intervals distinguish demonstrated improvement from inconclusive effects
O4: practicality	End-to-end median/p95 latency, cold load, peak memory and setup reproducibility	Runnable documented demo on stated hardware; added checking cost reported against its factual benefit

Factual audit. Plan at least 100 held-out pairs with two independent annotators, subject to a pilot workload/power assessment. Establish reference facts from original images and available labels without predicted masks, then judge anonymised outputs. Label atomic claims supported, unsupported, contradicted or unverifiable; adjudicate disagreements and report agreement. Supported-claim precision uses all emitted factual claims as its denominator. Reference-fact recall measures observable, in-scope facts conveyed. Empty descriptions have zero recall and no defined precision, not perfect reliability. Report omission, abstention and false-change rates separately.

Ablations. Compare: (A) images-only caption; (B) predicted-mask conditioning; (C) B plus mask consistency; (D) C plus direct image-backed judgment; (E) C plus disagreement-triggered reinspection and decision policy. Hold detector and candidate captions fixed for C-E; compare targeted and full inspection at matched evidence budgets. Add an oracle-mask diagnostic, natural detector-error strata and separately labelled injected mask/claim errors. Compare rules with the decision model using identical evidence. Stress decision wording and option order. Nuisance tests cover brightness/colour, blur/compression and small misalignment with valid overlap defined.

Standard and proposed overall measures. BLEU-4, METEOR, ROUGE-L and CIDEr-D remain supplementary caption measures. Region-count MAE is distinct from building-count MAE, which requires instance annotations. We also plan an exploratory end-to-end grounded-change score combining localisation, correct object/action claims, retained coverage and robustness. Its formula, matching rules and weighting will be fixed on development data before final evaluation and validated against blinded human rankings. Until validated, it is not a primary success criterion and cannot replace the component metrics.

Decision rule. The research hypothesis is supported only if the proposed checker improves factual quality over the controlled baselines at comparable useful coverage; otherwise the result is inconclusive or negative. Arbitrary promises of a 10-point gain, 25% robustness improvement or five-second runtime are replaced by measured effects and explicit uncertainty.

10. Milestone Plan

M2, Oct 29: provenance, label/split audits and bounded subset. **M3, Nov 5:** justified architecture, contracts and end-to-end subset. **M4, Nov 19:** detector training, bounded adapter experiments and logs. **M5, Nov 26:** frozen ablations, factual audit and failures. **M6, Dec 10:** reproducible demo and documentation [25]. Individual contributions will be tracked per milestone. At M1, models, mixed-data training and reliability improvements remain proposed rather than measured results.

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